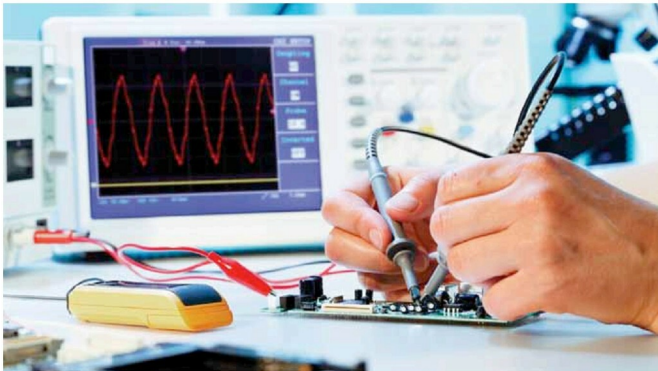


# Digital Storage Oscilloscopes

For serious electronics, a digital storage oscilloscope is the next must-buy tool after a DC variable power supply and digital multimeter. One can avoid an expensive mistake of buying a wrong oscilloscope by thoroughly understanding the specifications beforehand



## ANKIT GUPTA

An oscilloscope is a device that displays an electrical signal in the form of a graph. The graph in most cases shows how the signal changes over time. Y-axis represents voltage and X-axis represents time. These simple graphs can be used to extract highly useful information about the signals, such as amplitude, frequency, time period, rise time, duty cycle, jitter and much more.

Oscilloscopes can be classified as analogue and digital. Analogue oscilloscopes trace signals but digital oscilloscopes sample signals and construct waveforms on the display. Digital oscilloscopes use an analogue-to-digital converter (ADC) to convert the measured voltage into digital information.

They acquire the waveform as a series of samples. These samples are continuously stored in the memory until the oscilloscope has enough samples to describe a waveform. The digital oscilloscope then reproduces the waveform on the screen by using the stored samples.

Digital oscilloscopes are further classified into digital storage oscilloscopes (DSOs), digital phosphor oscilloscopes (DPOs), mixed-signal oscilloscopes (MSOs) and digital sampling oscilloscopes. Here we will discuss only DSOs. DSOs allow you to capture and view transients. Since the signal information exists in digital form as a series of stored digital codes, it can be processed easily by the oscilloscope itself or by an external computer to extract various results from

the measurements. The waveform can also be displayed even after the signal has disappeared.

## Why do you need an oscilloscope?

A good oscilloscope is an indispensable tool for anyone who is into design, testing or repair of electronic equipment. The oscilloscope helps you visualise the functioning of electronic circuits better. If you are into design, the detailed measurements can help you optimise your design for improved performance. In testing, a good insight into the circuit can be gained very quickly using an oscilloscope. Repairing becomes easy and quick with a tool to measure accurately at various testpoints. These days, engineers need the best tools available to solve their